



Hugo LERAY

Space Systems Engineer

Space System Engineer, currently working as Research Engineer on ionospheric characterization and space weather. Experienced in multi-receiver GNSS data analysis, EPB detection and velocity estimation using ML and signal processing methods. Skilled in satcom system design, deep learning signal analysis, Python/MATLAB comms-chain development and STK mission planning. Interested in R&D roles in NewSpace.

Education

● Master: Aeronautical and Space Systems Specialist Engineering program, English-speaking section

From Sept. 2020 to Feb. 2025 **Institut Polytechnique des Sciences Avancées (IPSA)** Ivry-sur-Seine, France

Space, Launchers and Satellite major.

Some projects :

- **Master Project:** Design of the electronic circuitry for the flight computer of a modular rocket probe. In collaboration with **Hybrogines**
- Autonomous exploration of robots using **Gazebo**
- Creation of a space mission to Neptune using **STK** and manual calculations
- **Machine learning** project of a character learning how to survive on an island using DEEP QL
- Creation of a full communication chain from a satellite to a receiver. Transmitter - Channel - Receiver using **Python**.

● Semester abroad

From Oct. 2023 to Feb. 2024 **Universiti Sains Malaysia** Penang, Malaysia

- Satellite communication
- Antenna and propagation
- Information system theory and management

Work experience

● Research Engineer - Promoted from Intern (April 2025 - September 2025)

From Oct. 2025 to Sept. 2026 **National University of Singapore** Singapore

Ionospheric topics, Equatorial Plasma Bubbles, GNSS signal analysis:

- Conduct research on equatorial plasma bubbles and ionospheric irregularities
- Perform multi-year GNSS data analysis to uncover novel trends and findings
- Analyze signal degradation and scintillation effects
- Develop and implement new methodologies to characterize the phenomenon using Machine Learning and other methodologies
- Contribute to scientific papers and conference publications
- Collaborate with researchers on data processing and interpretation

● Electromechanical technician internship

From Jun. 2024 to Aug. 2024 **Parker Meggitt Avrillé (Artus SAS)** Avrillé, France

Internship in the IPSA curriculum. Building mechanical, micromechanical, and electromechanical motors and gear reducers for the aerospace and defense industry.

Publications & Conferences

EuCAP 2026 (Dublin, Ireland) - Automatic Detection of Equatorial Plasma Bubbles Using Rate of TEC Index from SiReNT GNSS Data

- Neyman-Pearson detector with deep learning signal decomposition (sum of Gaussians)
- EPB clustering via IoU on Gaussian envelopes

AP/S-URSI 2026 (Detroit, USA) - Estimation of Equatorial Plasma Bubble Drift Velocities Using Dynamic Time Warping

- Constrained DTW alignment across receivers; drift velocity from planar wavefront model
- Validated on synthetic EPB simulator

URSI-GASS 2026 (Krakow, Poland) - Equatorial Plasma Bubble Velocity Estimation over Southeast Asia

- DTW-based velocity framework on SiReNT data with spatial interpolation

APCAP 2026 (Chengdu, China) - Occurrence and Characteristics of Post-Sunset Equatorial Plasma Bubbles over Singapore

- 2 Years analysis of results from our Automatic Detection of Equatorial Plasma Bubbles Using Rate of TEC Index from SiReNT GNSS Data

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Language

French

English

TOEIC: 27th May 2024

› TOEIC : 965/990

Computer skills

Python

Pytorch, Tensorflow

C/C++

Database & SQL

Matlab/Simulink

STK 12

FPGA (Quartus)

IDM

Pack Office

Tosa Excel 2019 : 797

CATIA V5 : 3D Modeling

Skills

Teamwork

Sense of initiatives and responsibility

Strengths

Willing, Helpful, Persevering

Travel

Austria, Morocco, Spain, England, Germany, Martinique, Guadeloupe, Greece, Thailand, Laos (alone), Malaysia, Singapore, Philippines, Indonesia, Vietnam